

Analogy between Signals and Vectors

- Signal space concepts and geometric representation of signals are very useful in the analysis of various digital modulation schemes and in statistical detection

Essential in vector space

Vector space , like 'Euclidian space' is a set of vectors, satisfy certain conditions

1. Sum of any two vectors belonging to V is another vector which also belongs to V
2. There exists in V , a vector, called zero vector, denoted by 0 , such that $A + 0 = A$, for any vector A belonging to V
3. For any vector A belonging to V , there exists another vector B , also belonging to V and such that $A + B = \text{zero vector } 0$
4. If any vector , Say A belonging to V is multiplied by a scalar , resultant vector also belongs to V

Vector A, such as shown in adjacent figure, can be expressed as

$$A = i x_1 + j x_2 + k x_3$$

Where i, j, k are unit vectors along X, Y, Z directions respectively.

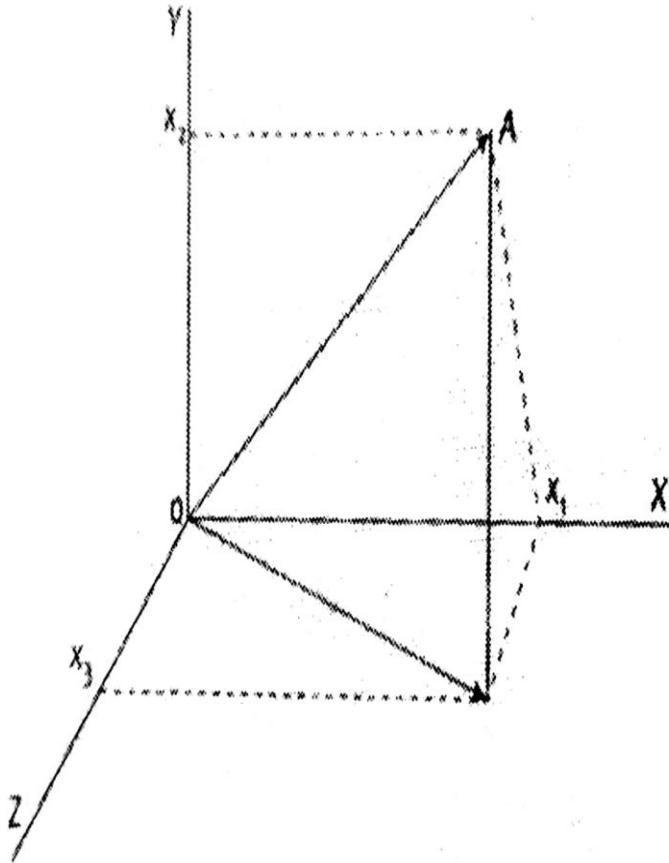
x_1, x_2, x_3 are real numbers representing co ordinates of vector A along X, Y and Z directions.

Vector A is represented as $A = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Unit vector i, j, k is represented as

$$i = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad j = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad k = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$



Linearly Independent vectors

- Vectors i, j, k are said to be linearly independent if none of them can be expressed as a linear combination of the rest
- Show that following three vectors belonging to the Euclidean space are linearly independent

$$X = (1, 6, 5) \quad Y = (1, 1, 0) \quad \text{and} \quad Z = (7, 5, 2)$$

Check determinant of matrix is zero or not.

$$A = \begin{bmatrix} 1 & 6 & 5 \\ 1 & 1 & 0 \\ 7 & 5 & 2 \end{bmatrix} = -20 ; \quad \text{matrix } A \neq 0 ;$$

So given vectors are linearly independent

Basis Vectors

- A set of vectors belonging to vector space V , which are linearly independent and which span that space are said to be forming a basis set for that space.
- Thus i, j, k form basis set of vectors
- Vectors i, j, k are not only linearly independent, also mutually orthogonal
- Two vectors are orthogonal, if their dot product is zero

Two vectors $x = (0,2,1)$ and $Y = (1, -2,1)$. Find magnitude of each vector and check whether they are orthogonal. If they are not orthogonal find angle between them.

- $A \bullet B = |A| |B| \cos \Theta$
- $|X| = \sqrt{0 + 4 + 1} = \sqrt{5}$
- $|Y| = \sqrt{6}$
- $X \bullet Y = 0 - 4 + 1 = -3 \neq 0$ X and Y are not orthogonal
- $\Theta = \cos^{-1} \left(\frac{X \bullet Y}{|X||Y|} \right) = \cos^{-1} \left(\frac{-3}{\sqrt{30}} \right) = \cos^{-1} (- \sqrt{0.3})$

- **Orthogonality of signals** $S_1(t)$ and $S_2(t)$ are orthogonal to each other over an interval t_1 to t_2 is zero.

Norm or length of signal

- Inner product of signal with itself , $\| S_1(t) \|^2 = E = \text{Energy of signal } S_1(t)$ over the interval t_1 to t_2 .
- Norm or length of finite energy of signal = $\sqrt{E}$

Distance between two signals

- Norm of their difference
- Distance between $S_1(t)$ and $S_2(t)$ = $\| S_1(t) - S_2(t) \|$
$$= \left[\int_{t_1}^{t_2} |S_1(t) - S_2(t)|^2 dt \right]^{1/2}$$

= square root of energy of the difference signal over the interval t_1 to t_2